

Web Console (no tunnel)

Web Console (no tunnel)

Revision: 2 Last modified: 2026-07-04T12:00:00Z

Master technical specification — Volume 4 (Clients), nano-detail deep-dive. This document **deepens** the *Helix Console builds to Web (API-only, no core_ffi)* and the *optional in-page WASM MASQUE proxy* parts of the pass-1 client overview [04_UI §5.7, §6, 03-CLIENT §5.7/§6/CI3] into an implementation-ready specification of the **Web flavor**: the build-time exclusion of every tunnel surface, the NoCore null-object contract, the responsive admin UI, browser auth (OIDC + anonymous device-token), the `helix_api` REST + WS/SSE consumption with its reconnect state machine, and the **browser-scoped WASM MASQUE proxy** (explicitly *not* a system VPN). SPEC-ONLY: it describes **what to build**, not the shipping product.

Boundary with sibling docs. This document **consumes**: the FFI surface + TunnelStatus enum owned by 03-client-core-and-ui.md §3 [03-CLIENT §3]; the orchestrator TunnelStatus source-of-truth in v02-data-plane/ orchestrator-and-state.md §4.1 [ORCH §4.1]; the Transport trait owned by v02-data-plane/transport-trait.md [TX-TRAIT]; the control-plane REST/WS contract owned by doc 02. It **owns**: the Web-flavor build manifest, the NoCore/no-TunnelPlatform contract, the Console session + WS-reconnect state machines, browser auth lifecycle, and the WASM MASQUE proxy contract.

Evidence base, cited inline by id. [04_UI §N] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [03-CLIENT §N] = final/03-client-core-and-ui.md; [ORCH §N] = v02-data-plane/orchestrator-and-state.md; [TX-TRAIT] = v02-data-plane/transport-trait.md; [SYN §N] = final/.../v09-research/_SYNTHESIS.md; [research-flutter_ffi], [research-ios_android]

= cited research digests; [05_YB0] = operator mandate. Any claim not grounded in the evidence base is marked **UNVERIFIED** per constitution §11.4.6.

Table of contents

- [0. Position, ownership, invariants](#)
- [1. The structural fact: no system TUN in a browser](#)
- [2. The Web-flavor build manifest \(what compiles, what is excluded\)](#)
- [3. The NoCore null-object contract](#)
- [4. Console capability gating + runHelixApp web path](#)
- [5. Responsive admin UI](#)
- [6. Auth — OIDC + anonymous device-token \(browser lifecycle\)](#)
- [7. helix_api consumption — REST + WS/SSE + reconnect state machine](#)
- [8. The optional in-page WASM MASQUE proxy \(caveat surface\)](#)
- [9. Memory / size / performance budgets \(Web\)](#)
- [10. Error handling & edge cases](#)
- [11. Test points \(mapped to §11.4.169 test-type set\)](#)
- [12. Phase → task → subtask plan](#)
- [13. Open decisions surfaced by this document](#)
- [14. Cross-document contracts this document fixes](#)
- [Sources verified](#)

0. Position, ownership, invariants

The **Web flavor is the Helix Console only** — the admin app (tenants, users, devices, networks, routes, policies, audit, billing-optional) [04_ARCH §2 roles, 04_UI §8.3]. Access and Connector **do not** build to Web, because a browser cannot open a system TUN device (§1). The Web flavor is the *single* flavor that omits `helix_core_ffi` and ships **no TunnelPlatform implementation** [03-CLIENT CI3, 04_UI §1/§5.7]. Optionally, a Console page may host a **browser-scoped WASM MASQUE proxy** that tunnels *only the page/browser's own* fetch traffic — never the OS (§8) [04_ARCH §5.7].

0.1 What this document owns

#	Contract	§
C1	Web-flavor build manifest — the exact set of packages that compile (<code>helix_design</code> , <code>helix_api</code> , <code>helix_domain</code> , <code>helix_ll0n</code>) and the exact set excluded (<code>helix_core_ffi</code> , every <code>TunnelPlatform</code> shim)	§2

#	Contract	§
C2	NoCore null-object — the compile-time stand-in for HelixCore when Capability.tunnel is absent; every method throws UnsupportedOnWeb, never silently no-ops	§3
C3	Console session + auth lifecycle — OIDC code-flow (PKCE) + anonymous device-token, token storage, refresh, logout, the session state machine	§6
C4	helix_api consumption + WS/SSE reconnect state machine — the reactive admin data layer driven by the control-plane event stream	§7
C5	WASM MASQUE proxy contract — WasmMasqueProxy Dart facade + the wasm-bindgen exports it drives; its own ProxyStatus enum (distinct from TunnelStatus); the honest "not a VPN" boundary	§8

0.2 What this document does NOT own

- The REST/WS schema, WatchNetworkMap, enrollment, device.revoked taxonomy — doc 02 (this doc *consumes* the generated helix_api client).
- The TunnelStatus/Shields/ExitOption FFI types — 03-CLIENT §3 (Console links none of them; the WASM proxy defines its *own* status type, §8.4).
- The Transport trait, MASQUE/quinn+h3 internals — [TX-TRAIT] / doc 01 (the WASM proxy compiles a *subset* of helix-transport to wasm32, §8.2).
- helix_design tokens / signature components — 03-CLIENT §7 (Console reuses them; the connection-state palette is reused for the proxy chip only).

0.3 Invariants this document inherits and tightens

#	Invariant	Source	Web-Console tightening
WI1	Console depends on helix_api only — never helix_core_fffi; this is what lets it build to Web.	[03-CLIENT CI3, 04_UI §1]	§2: a build that links helix_core_fffi into the web target is a build-blocking drift finding (§2.3).
WI2	No system tunnel in a browser — structural, not a missing feature (§11.4.112).	[04_ARCH §5.7]	§1: stated to the user in-product; the WASM proxy is page-scoped only.
WI3	UI is a pure function of a stream — Console	[03-CLIENT]	§7: the same honesty rule applied to admin data; stale data is

#	Invariant	Source	Web-Console tightening
	folds the WS/SSE event stream, not polling.	CI2, 04_UI §4.3]	<i>labelled</i> stale, never silently shown as live.
WI4	State announced, not just colored — ally + safety.	[03-CLIENT CI5, 04_UI §9]	§5/§8.4: the proxy on/off and every "online/offline" badge carry a semantic label + icon, not color alone.
WI5	The WASM proxy is browser-scoped, evidence-honest — it MUST NOT claim to protect the device.	[04_ARCH §5.7]	§8.5: the proxy UI text + the ProxyStatus enum forbid any "device protected" wording.
WI6	No secrets in JS-reachable storage beyond what the flow requires; no durable connection log.	[SYN §7]	§6.4: tokens in memory + sessionStorage (not localStorage) by default; the WASM proxy keeps aggregate counters only (§8.6).

[MERMAID RENDER FAILED — block #1]

1. The structural fact: no system TUN in a browser

A browser process **cannot** create or own a TUN/utun device, install routes, or set system DNS — there is no Web API for it, by sandbox design [04_ARCH §5.7, 03-CLIENT §5.7]. Per constitution §11.4.112, this is a **structural impossibility**, not unimplemented work: the Console therefore ships **no TunnelPlatform implementation and no helix_core_ffi dependency** on Web. The in-product copy states this plainly (WI2/WI5): *"On the web, Helix Console manages your networks. To protect this device, install the Helix Access app."*

What a browser **can** do (and only this): open **MASQUE CONNECT-UDP over HTTP/3** sockets from page JavaScript/WASM to the gateway edge, and rewrite the **page's own** outbound fetch/XMLHttpRequest through them — a *browser-scoped proxy*, not a device VPN (§8). The boundary is load-bearing and never blurred.

Honest boundary (§11.4.6, §11.4.112). "Fully responsive web app" [05_YBO] = the Console **plus** an optional browser-scoped proxy. It is **never** a system-wide tunnel. A reopen of this verdict requires NEW evidence a browser gained TUN access (§11.4.7 / §11.4.34) — not a re-derivation.

2. The Web-flavor build manifest (what compiles, what is excluded)

2.1 Package set (melos workspace, web target) [04_UI §1, 03-CLIENT §2]

```
# apps/console/pubspec.yaml – the Web flavor's dependency closure
name: helix_console
dependencies:
  helix_design: { path: ../../packages/
                  helix_design } # tokens, theme, widgets, ally
  helix_api:    { path: ../../packages/helix_api } #
                  generated REST + WS/SSE client (doc 02)
  helix_domain: { path: ../../packages/helix_domain } #
                  providers, usecases, runHelixApp, NoCore
  helix_l10n:   { path: ../../packages/helix_l10n } # ARB
                  localization
  flutter_riverpod: ^2.x
#
  NOTE: helix_core_ffi is DELIBERATELY ABSENT (W11). It is
        NOT a transitive dep of any of the above.
flutter:
  uses-material-design: true
```

helix_domain MUST be authored so that **none** of its web-reachable code paths import helix_core_ffi. The dependency is provided *only* to Access/Connector; helix_domain references the core through an **interface** (HelixCore, §3) that the Web build satisfies with NoCore. This is the standard null-object/dependency-inversion pattern [research-flutter_ffi].

2.2 Conditional import seam (so one helix_domain serves all flavors)

```
// helix_domain/lib/src/core/core_factory.dart
// Dart conditional imports pick the platform-correct factory at
// compile time.
import 'core_factory_stub.dart'
  if (dart.library.io) 'core_factory_io.dart' // mobile/
  desktop → RealHelixCore (links core_ffi)
  if (dart.library.js_interop)
    'core_factory_web.dart'; // web → NoCore (never references
    core_ffi)
```

```

abstract interface class CoreFactory {
  HelixCore create(Set<Capability> caps);
}

// helix_domain/lib/src/core/core_factory_web.dart - the WEB
// compilation unit
// CRITICAL: this file MUST NOT import helix_core_ffi (would pull
// the staticlib into the web build).
import 'no_core.dart';
class CoreFactoryWeb implements CoreFactory {
  @override HelixCore create(Set<Capability> caps) => const
    NoCore();
}

```

Why a conditional import and not a runtime if: a runtime branch still *links* `helix_core_ffi` into the web bundle (dead weight + a `dart:ffi` reference that does not compile to Wasm/JS). The `if (dart.library.js_interop)` seam means the web compiler never sees the FFI unit at all [research-flutter_ffi].

2.3 Drift gate (the no-FFI-on-web guarantee)

A pre-build gate CM-WEB-NO-CORE-FFI (§11.4.110 clash-detector class) asserts: the flutter build web dependency graph for apps/console contains **zero** package:helix_core_ffi/* imports and zero dart:ffi references. A violation is a **build-blocking** finding (WI1). Paired §1.1 mutation: add a `helix_core_ffi` import to a console-reachable file → gate FAILs (§11). This is the mechanical form of CI3.

2.4 Web build configuration

Concern	Value	Rationale
Renderer	CanvasKit (default) for fidelity; Wasm/-wasm when the toolchain target supports it for the heavy admin tables	[04_UI §10.web, 04_ARCH §5.6]
Routing	URL-strategy = path (clean URLs); route-level deferred loading of heavy admin modules (deferred as topology)	[04_UI §7]
CSP	strict Content-Security-Policy; the WASM proxy module loaded under a nonce'd script-src + wasm-unsafe-eval only when the proxy capability is enabled (§8.3)	security

Concern	Value	Rationale
Base href	configurable for sub-path hosting behind the gateway's reverse proxy	deploy flexibility

3. The NoCore null-object contract

helix_domain references the tunnel core through the HelixCore interface [03-CLIENT §3.2]. On Web, Capability.tunnel is absent, so the provider is overridden with NoCore — a null object whose every method **throws a typed, honest error**, never silently succeeds (WI2; §11.4.6 no-bluff: a no-op that returned a fake Connected would be a PASS-bluff).

```
// helix_domain/lib/src/core/no_core.dart
import 'package:helix_domain/src/core/helix_core.dart'; // the
    interface ONLY (no ffi)

/// Thrown by every NoCore method. The UI must never display
    "connected" on web.
final class UnsupportedOnWeb implements Exception {
  final String op;
  const UnsupportedOnWeb(this.op);
  @override String toString() =>

    'UnsupportedOnWeb: "$op" – Helix Console (web) has no
    system tunnel; '
    'install Helix Access to protect this device. (WI2,
    §11.4.112)';
}

final class NoCore implements HelixCore {
  const NoCore();

  // status stream is a SINGLE-element stream emitting the honest
    terminal state, then closes.
  // It never emits Connecting/Connected – the web build has no
    tunnel to report (WI2).
  @override
  Stream<TunnelStatus> statusStream() =>
    Stream.value(const TunnelStatus.disconnected()); // const,
    never updated

  @override Future<void> start({required String transport,
    String? mapPathOrSession,
    CoreMode mode = CoreMode.client}) async => throw const
    UnsupportedOnWeb('start');
  @override Future<void> stop() async => throw const
    UnsupportedOnWeb('stop');
  @override Future<List<ExitOption>> exits() async => throw const
    UnsupportedOnWeb('exits');
```

```

@override Future<void> setExit(String id, {List<String>?
    multiHopChain}) async
    => throw const UnsupportedOnWeb('setExit');
@override Future<void> setShields(Shields s) async => throw
    const UnsupportedOnWeb('setShields');
@override Future<AdvertiseResult> advertise(List<String> cidrs)
    async
    => throw const UnsupportedOnWeb('advertise');
@override Future<void> attachTun(int fd) async => throw const
    UnsupportedOnWeb('attachTun');
@override Future<void> detachTun() async => throw const
    UnsupportedOnWeb('detachTun');
}

```

TunnelStatus consistency note. The Dart TunnelStatus here is the *same* generated mirror Access/Connector use [03-CLIENT §3.1] — Console links the **type** (a tiny generated Dart enum, no Rust staticlib) but never the tunnel implementation. The canonical Rust source is the orchestrator's 5-variant enum Connecting | Handshaking | Connected{transport,rtt_ms} | Reconnecting | Down{reason} [ORCH §4.1], which 03-CLIENT §3.1 extends to 7 variants (Disconnected, Connected{...,path}, Danger{kind}). NoCore only ever uses Disconnected. The byte-for-byte FFI contract test [03-CLIENT §12] still applies to the *type definition* even on web.

The Console UI **never** renders a ConnectButton/ShieldIndicator (Capability.tunnel tree-shakes them out, §4); NoCore exists purely so that any accidental call surfaces as a loud typed error in tests rather than a silent fake-success.

4. Console capability gating + runHelixApp web path

The Console entrypoint sets capabilities: {Capability.admin} only — no Capability.tunnel, so helixCoreProvider resolves to NoCore and every tunnel widget is tree-shaken [03-CLIENT §6, 04_UI §2.console].

```

// apps/console/lib/main_console.dart
void main() => runHelixApp(
    flavor: HelixFlavor.console,
    home: const ConsoleShell(), // tenants/devices/
    networks/policy/audit/topology
    capabilities: const {Capability.admin}, // NO
    Capability.tunnel → NoCore, no core_ffi (CI3)
);

```

```
// helix_domain/lib/run_helix_app.dart (the web branch of the
// shared wiring point)
Future<void> runHelixApp({
  required HelixFlavor flavor,
  required Widget home,
  required Set<Capability> capabilities,
}) async {
  final factory = const CoreFactory.platform(); // §2.2
  // conditional import → CoreFactoryWeb on web
  final container = ProviderContainer(overrides: [
    flavorProvider.overrideWithValue(flavor),
    capabilitiesProvider.overrideWithValue(capabilities),
    // web: factory.create({admin}) returns NoCore (no
    // Capability.tunnel) – never a RealHelixCore.
    helixCoreProvider.overrideWith((ref) =>
      factory.create(capabilities)),
    // tunnelPlatformProvider is NULL on web – there is no
    // TunnelPlatform impl (WI1):
    tunnelPlatformProvider.overrideWith((ref) =>
      capabilities.contains(Capability.tunnel) ? resolveShim() :
      const NoTunnelPlatform()),
  ]);
  runApp(UncontrolledProviderScope(container: container,
    child: HelixApp(home: home, theme:
      helixTheme(flavor)))); // Console theme (§5)
}

// helix_domain/lib/src/core/no_tunnel_platform.dart – web has no
// shim (WI1)
final class NoTunnelPlatform implements TunnelPlatform {
  const NoTunnelPlatform();
  @override Future<void> startTunnel(TunnelConfig cfg) async =>
    throw const UnsupportedOnWeb('startTunnel');
  @override Future<void> stopTunnel() async => throw const
    UnsupportedOnWeb('stopTunnel');
  @override Stream<PlatformTunnelEvent> events() => const
    Stream.empty();
}
```

Capability	Compiles in (web)	Notes
admin	tenants/devices/networks/ policy/audit/topology screens + helix_api heavy tables	the only Console capability [03- CLIENT §6]
tunnel	absent → ConnectButton, ShieldIndicator, ExitPicker, helix_core_ffl, every shim tree- shaken	CI3 / WI1
proxyWasm (optional, §8)	the WASM MASQUE proxy module + its on/off chip;	Phase 3; never implies a device tunnel

Capability	Compiles in (web)	Notes
	deferred-loaded; gated behind a build flag + CSP	

5. Responsive admin UI

The Console reuses `helix_design` [03-CLIENT §7] and renders an `AdaptiveScaffold` that reflows on **width**, never on `Platform.isX`, so a resized desktop window behaves like a phone and "web just works" [04_UI §7].

// breakpoints (identical to the shared rule, 03-CLIENT §7.3) – branch on SIZE

```
compact  (< 600)    → BottomNavigationBar · single pane
medium   (600–1024) → NavigationRail · master/detail
expanded (> 1024)   → extended NavigationRail · multi-pane (list
                    | detail | live events)
```

5.1 Console shell + input model

- **ConsoleShell** hosts the `NavigationRail` (`Tenants · Devices · Networks · Policy · Audit · Topology`) and a multi-pane body on expanded.
- **Pointer + keyboard first** on desktop/web: `⌘K/Ctrl-K` command palette (jump-to-device, jump-to-network, run-policy-diff), full keyboard nav, focus rings [04_UI §7, §9].
- **Deferred loading** of the heavy modules (`PolicyEditor`, topology graph, audit table) via deferred as imports so first paint is fast (§9) [04_UI §7].
- **Live, never polled** (WI3): device lists / topology / audit fold the WS/SSE stream (§7) — a refresh button forces a snapshot re-fetch but the steady state is push-driven.

5.2 Signature components reused

Component	Console use	Driven by
NetworkTile	a joined network row (connector health, advertised CIDRs, access level)	<code>helix_api</code> + WS <code>route.changed</code>
PolicyEditor	ACL editing with effect-diff preview + version history + one-click rollback	<code>helix_api</code> policy endpoints (doc 02)
AdaptiveScaffold	the responsive shell	layout width
StatusChip		<code>ProxyStatus</code> (§8.4)

Component	Console use	Driven by
	reused only by the optional WASM proxy chip (§8.4) — NOT a tunnel status	

5.3 Accessibility (WI4, CI5)

Every "online/offline / healthy/degraded / protected-page/unprotected" indicator carries a **semantic label + icon shape**, never color alone [04_UI §9]: e.g. a degraded connector is announced "*warehouse-gw: degraded*" with a warning glyph, not just amber. Light + dark are both first-class (admins skew dark) [04_UI §3.1].

6. Auth — OIDC + anonymous device-token (browser lifecycle)

The Console authenticates the **admin user** (not a device tunnel). Two paths [04_ARCH §4.2 users, 04_UI §8.2]: **OIDC** (managed orgs) and **anonymous device-token** (self-host / homelab, no PII). The browser context constrains the token handling more tightly than the native apps (XSS surface), so this section fixes the web-specific lifecycle.

6.1 OIDC authorization-code flow with PKCE (browser-correct)

[MERMAID RENDER FAILED — block #2]

- **PKCE (S256) is mandatory** for the browser (public client) — no client secret ships in JS [research-flutter_ffi: SPA auth best practice; **UNVERIFIED** exact IdP product, gated on the §02 identity choice].
- **Token exchange goes through the Helix backend** (POST / v1/auth/token), not the SPA directly, so the refresh token can live in an **httpOnly, Secure, SameSite=Strict cookie** the JS never reads (XSS mitigation).
- state is verified on callback (CSRF); nonce echoed if the IdP returns an id_token.

6.2 Anonymous device-token (self-host)

For homelab/self-host with no IdP, the admin presents a **bootstrap admin token** minted by `helixvpctl init [04_ARCH $10]` (or a QR/enroll-token). The Console exchanges it once at `POST /v1/auth/token {device_token}` for the same short-lived session shape. No email/PII is stored (WI6) [04_UI §8.2].

6.3 Session state machine

[MERMAID RENDER FAILED — block #3]

6.4 Token storage policy (the web-specific hardening, WI6)

Material	Storage	Rationale
access_token (short TTL)	in-memory only (Riverpod state)	not readable after tab close; minimal XSS exfil window
refresh_token	httpOnly Secure SameSite=Strict cookie (server-set)	JS cannot read it (XSS)
code_verifier, state, nonce	sessionStorage (cleared on tab close)	flow-scoped, single- use
PII (anonymous mode)	none	WI6, §11.4.10

`localStorage` is **never** used for tokens (persists across XSS, survives tab close). On 401 from any REST/WS call, the session machine transitions to Refreshing; a failed refresh forces Unauthenticated and a re-login [research-flutter_ffi].

UNVERIFIED: whether the deployment fronts the Console with the gateway reverse proxy doing the cookie handling vs the Go API directly — recorded as a follow-up (§13 D-WEB-2), default = behind the gateway.

6.5 Auth controller (Dart, Riverpod)

```
// helix_domain/lib/src/auth/auth_controller.dart
final authProvider = AsyncNotifierProvider<AuthController,
    Session>(AuthController.new);

class AuthController extends AsyncNotifier<Session> {
  @override Future<Session> build() async =>
    Session.unauthenticated();

  Future<void> beginOidc() async { /* PKCE: store verifier,
    redirect to /authorize */ }
```

```

Future<void> completeOidc(Uri callback) async {
  _verifyState(callback); //
  CSRF (§6.1)
  final code = callback.queryParameters['code']!;
  final tok = await ref.read(authApiProvider).exchange(code:
    code, verifier: _popVerifier());
  state = AsyncData(Session.authenticated(tok));
  _scheduleRefresh(tok.expiresIn); //
  §6.3 Refreshing
}
Future<void> loginWithDeviceToken(String t) async { /* POST /v1/
  auth/token {device_token} */ }
Future<void> refresh() async { /* uses httpOnly cookie; on fail
  → unauthenticated */ }
Future<void> logout() async { /* clear memory + POST /v1/auth/
  logout (clears cookie) */ }
}

```

7. helix_api consumption — REST + WS/SSE + reconnect state machine

The Console's data layer is helix_api (generated OpenAPI REST + a WS/SSE client) folded into Riverpod, exactly the [03-CLIENT §8.3] Console live-data pattern. This document fixes the **web reconnect state machine** (the browser's WebSocket lifecycle differs from native: bfcache freeze, tab visibility, network-change events).

7.1 REST layer

helix_api is the generated client; every call carries the Authorization: Bearer <access_token> from §6 and surfaces typed errors. CRUD on tenants/users/devices/networks/routes/policies/audit [04_ARCH §6, 04_UI §8.3].

```

final devicesProvider = FutureProvider<List<Device>>((ref) async {
  final api = ref.watch(restClientProvider); //
  generated, auth-injected
  return api.listDevices(tenantId:
    ref.watch(currentTenantProvider));
});

```

7.2 WS/SSE live event fold (push, never poll — WI3)

```

// helix_domain/lib/src/live/event_stream_provider.dart
final eventStreamProvider = StreamProvider<ControlEvent>((ref) {
  final ws = ref.watch(wsClientProvider); //
  GET /v1/stream (doc 02)
}

```

```

    return ws.events(); //
        device.online | route.changed | policy.compiled |
        device.revoked
});

// folded views update live:
final deviceListProvider = NotifierProvider<DeviceListNotifier,
    List<Device>>(DeviceListNotifier.new);
class DeviceListNotifier extends Notifier<List<Device>> {
    @override List<Device> build() {
        ref.listen(eventStreamProvider, (_, ev) =>
            ev.whenData(_apply)); // fold events into the list
        return ref.watch(devicesProvider).valueOrNull ?? const
            []; // seed from REST snapshot
    }
    void _apply(ControlEvent e) { /* online/offline/route/policy/
        revoke → mutate state */ }
}

```

The Console event taxonomy is doc 02's Phase-1 set (device.online, route.changed, policy.compiled, device.revoked) made visible [04_UI §4.3].

7.3 WS reconnect state machine (browser lifecycle aware)

[MERMAID RENDER FAILED — block #4]

Trigger	Handling	Why
socket close / error	enter Backoff; jittered exp (base 250 ms, cap 30 s, full jitter) — mirrors the data-plane backoff shape [ORCH §7.2]	avoid thundering-herd on a gateway restart
navigator.onLine = false	pause re-dials until online event	browsers fire offline/online; don't burn retries
visibilitychange → hidden / bfcache	enter Frozen; do not mark data stale yet (a hidden tab is normal)	UX
visibilitychange → visible	Resyncing: re-open WS and re-fetch the REST snapshot to heal any events missed while frozen	correctness — WS gaps are not replayed
401 on WS upgrade	hand to §6.3 Refreshing, then retry	token expiry mid-session

Staleness honesty (WI3, §11.4.6). While in Backoff/Frozen/Resyncing, the affected panes show a **"reconnecting... last updated HH:MM:SS"** banner — the UI never paints stale admin data as live. A device shown "online" whose last event is older than the staleness budget is labelled, not silently trusted (the admin-data analogue of the §8.4 Access honesty rule).

8. The optional in-page WASM MASQUE proxy (caveat surface)

A Console page **may** host a **browser-scoped MASQUE proxy** that tunnels *only the page's / browser's own* outbound HTTP(S) to a joined network — **not** a system VPN [04_ARCH §5.7, 03-CLIENT §5.7]. This is a Phase-3, opt-in, clearly labelled feature. Its honesty is constitutional (WI5; §11.4.112): the UI and the type system both forbid any "device protected" claim.

8.1 What it is and is NOT

It IS	It is NOT
a page-scoped proxy: the Console's own fetch/XHR (and optionally same-origin app traffic) routed via MASQUE CONNECT-UDP/H3 to the gateway	a system TUN / OS VPN
a reuse of the helix-transport MASQUE client compiled to wasm32-unknown-unknown	a re-implementation of WG/transport in JS
evidence-honest: a ProxyStatus chip, aggregate counters only	a leak-proof device kill-switch (browser cannot enforce that)

8.2 The WASM module (helix-transport subset → wasm32)

The proxy compiles the **MASQUE CONNECT-UDP** path of helix-transport [TX-TRAIT, 04_ARCH §3.3] to WebAssembly via wasm-bindgen. It uses the browser's **WebTransport** API (HTTP/3 datagrams) as the carrier where available; **UNVERIFIED**: the exact browser WebTransport ↔ MASQUE CONNECT-UDP mapping is recorded as a Phase-3 research item (§13 D-WEB-1) — the spike must prove a browser can drive the gateway's :443/udp MASQUE endpoint, or the feature degrades to "not available in this browser" (honest SKIP, never a fake proxy).

```

// helix-transport/src/wasm/masque_proxy.rs - wasm-bindgen
// exported surface (browser-only build)
// Compiled with: wasm-pack build --target web --no-default-
// features --features "masque,wasm"
use wasm_bindgen::prelude::*;

#[wasm_bindgen]
pub struct WasmMasqueProxy { /* opaque: quinn-over-WebTransport
    session + CONNECT-UDP stream */ }

#[wasm_bindgen]
impl WasmMasqueProxy {
    /// gateway_url = "https://gw.example:443" ; session = the §6
    /// Console session token.
    #[wasm_bindgen(constructor)]
    pub fn new(gateway_url: String, session_token: String) ->
        WasmMasqueProxy { /* ... */ }

    /// Open the MASQUE CONNECT-UDP session. Resolves when the H3
    /// datagram flow is up.
    #[wasm_bindgen]
    pub async fn start(&self) -> Result<(),
        JsValue>; // Err = honest failure (no fake "up")

    /// Tear down the session; flushes counters. Idempotent.
    #[wasm_bindgen]
    pub async fn stop(&self) -> Result<(), JsValue>;

    /// Proxy ONE outbound request's bytes through the tunnel;
    /// returns the response bytes.
    /// The page's fetch hook (§8.3) calls this. Browser-scoped
    /// only.
    #[wasm_bindgen]
    pub async fn proxy_request(&self, req: web_sys::Request) ->
        Result<web_sys::Response, JsValue>;

    /// JSON status snapshot: { state, carrier, rtt_ms, bytes_in,
    /// bytes_out } – aggregate only (§8.6).
    #[wasm_bindgen]
    pub fn status(&self) -> JsValue;
}

```

8.3 Dart facade + fetch interception

```

// apps/console/lib/proxy/wasm_masque_proxy.dart (loaded ONLY
// when Capability.proxyWasm, §4)
import 'dart:js_interop';

@JS('WasmMasqueProxy')
extension type _JsProxy._(JSObject _) implements JSObject {
    external _JsProxy(String gatewayUrl, String sessionToken);
    external JSPromise start();
    external JSPromise stop();
}

```

```

    external JSPromise<JSObject> proxyRequest(JSObject request);
    external JSObject status();
}

final class WasmMasqueProxy {
    final _JsProxy _p;
    WasmMasqueProxy(String gw, String session) : _p = _JsProxy(gw,
        session);

    Future<void> start() async => (await _p.start().toDart);    //
        throws on failure (no fake up)
    Future<void> stop() async => (await _p.stop().toDart);
    ProxyStatus status() =>
        ProxyStatus.fromJson(_p.status().dartify()! as Map);

    /// Install a fetch hook so the PAGE's requests go through the
    /// tunnel. Browser-scoped (WI5).
    void installFetchHook() { /* monkey-patch window.fetch →
        _p.proxyRequest, page scope only */ }
    void removeFetchHook() { /* restore original fetch on stop */ }
}

```

- The WASM module + its wasm-unsafe-eval CSP allowance load **only** when the proxyWasm capability is on (§4) and behind a nonce'd script-src (§2.4).
- The fetch hook is **page-scoped**: it rewrites window.fetch for the Console tab only. It does **not** and **cannot** intercept other tabs, other apps, or the OS (WI5).

8.4 ProxyStatus — a DISTINCT type from TunnelStatus

The proxy MUST NOT reuse TunnelStatus (that type implies a *device* tunnel and drives the stateConnected "you are protected" palette). It defines its own:

```

// apps/console/lib/proxy/proxy_status.dart
sealed class ProxyStatus {
    const ProxyStatus();
}

final class ProxyOff extends ProxyStatus { const
    ProxyOff(); }
final class ProxyStarting extends ProxyStatus { const
    ProxyStarting(); }
final class ProxyOn extends ProxyStatus
{
    // page-scoped, NOT a device VPN
    final String carrier; // "masque-h3"
    final int rttMs;
    const ProxyOn(this.carrier, this.rttMs);
}

final class ProxyUnavailable extends ProxyStatus
{
    // browser lacks WebTransport, etc.
    final String reason; // honest SKIP reason (§11.4.3)
    const ProxyUnavailable(this.reason);
}

```

```

}
final class ProxyError extends ProxyStatus { final String
    detail; const ProxyError(this.detail); }

```

The proxy chip label is explicit: **"Browser proxy: ON · masquerade · 23ms"** with a sub-line *"this tab only — not a device VPN"* (WI5). It reuses StatusChip (§5.2) and the stateConnecting/ stateConnected *amber/green motion* for ON, but never the word "protected" and never the full-screen ConnectButton.

8.5 The honesty boundary (WI5, §11.4.112)

Forbidden wording / behaviour	Required instead
"Your device is protected" / "VPN on"	"Browser proxy on (this tab only)"
a system-wide kill-switch claim	"browser-scoped; other apps are not tunnelled"
painting the device-Connected green full-bleed	the scoped chip only
silent fallback to no-proxy while showing "on"	ProxyUnavailable{reason} honest SKIP

8.6 No-logging by construction (WI6, I5)

The WASM proxy keeps **aggregate counters only** (bytes_in, bytes_out, rtt_ms EWMA, escalation count) — never per-request URLs, never durable records, mirroring the data-plane no-logging rule [SYN §7, ORCH §4.5]. status() exposes only those aggregates.

8.7 Memory (the browser-tab analogue of the iOS NE ceiling)

[MERMAID RENDER FAILED — block #5]

A browser tab has no hard ~15 MB ceiling like the iOS NE [03-CLIENT CI4], but the WASM heap is still budgeted: cap QUIC flow-control windows in the wasm build, bound the in-flight datagram buffers, and refuse (with ProxyUnavailable) rather than OOM the tab (§9). **UNVERIFIED:** measured WASM RSS under sustained transfer — a Phase-3 soak gate analogous to G3 (§11, §12).

9. Memory / size / performance budgets (Web)

Budget	Target	How
Initial bundle (Console, no proxy)	first paint fast; main bundle lean	route-level deferred loading of topology/policy/audit; tree-shake icons/fonts; no core_ffi weight (WI1) [04_UI §10.web]
Renderer	fidelity + speed	CanvasKit default; -wasm for heavy tables where supported (§2.4)
Heavy admin modules	never block first paint	deferred as imports; load on route entry
WASM proxy module	loaded only on proxyWasm opt-in	not in the base bundle; gated by CSP (§8.3)
Proxy WASM heap	bounded, refuse-not-OOM	capped QUIC windows + bounded datagram buffers (§8.7)
Live data	push, not poll	WS/SSE fold (§7); no polling loops burning the event loop (WI3)

Non-negotiable (CI6): never Electron, never webview-wrap — the Console is a real Flutter web app [04_UI §10, 03-CLIENT CI6]. The whole point of excluding core_ffi (WI1) is that the Web bundle stays lean precisely because it carries no tunnel core.

10. Error handling & edge cases

#	Edge case	Handling	Source
E1	A Capability.tunnel widget reached on web (bug)	NoCore throws UnsupportedOnWeb → loud test failure, never a fake Connected	§3, WI2
E2	helix_core_ffi accidentally imported into a console path	CM-WEB-NO-CORE-FFI build gate FAILs (§2.3)	§2.3, WI1
E3	OIDC state mismatch on callback	AuthError; discard code; do not exchange (CSRF defence)	§6.1
E4	access_token expiry mid-session		§6.3

#	Edge case	Handling	Source
		session machine Refreshing; transparent retry; failed refresh → re-login	
E5	WS dropped while tab hidden (bfcache)	Frozen → on visible Resyncing re-fetches REST snapshot to heal missed events	\$7.3
E6	navigator.onLine = false	pause re-dials; resume on online event (don't burn backoff)	\$7.3
E7	Browser lacks WebTransport / MASQUE path	proxy → ProxyUnavailable{reason} (honest SKIP), feature greyed out, never a fake proxy	\$8.2/ \$8.4
E8	Proxy WASM nears tab memory budget	refuse new flows with ProxyError/ProxyUnavailable, never OOM the tab	\$8.7
E9	401 from a REST call	route to §6.3 Refreshing; on fail force Unauthenticated	\$6.4
E10	Stale admin data during reconnect	"reconnecting... last updated HH:MM:SS" banner; never paint stale as live	\$7.3, WI3
E11	device.revoked event arrives	fold into device list live; if it is the <i>current admin's</i> session-tied device in anonymous mode, force re- auth	\$7.2
E12	Sub-path hosting behind gateway reverse proxy	configurable base href + cookie path scoping (§6.4)	\$2.4

11. Test points (mapped to §11.4.169 test-type set)

Constitution §11.4.169 mandates comprehensive coverage of a CLOSED test-type set, every PASS citing rock-solid captured **physical** evidence (§11.4.5/.69/.107), zero bluff; the only permitted absence is an honest §11.4.3 SKIP-with-reason. Web-Console coverage:

Test type (§11.4.169)	Web-Console coverage	Evidence
unit		test logs

Test type (§11.4.169)	Web-Console coverage	Evidence
integration (real System, infra via containers §11.4.76)	NoCore every method throws UnsupportedOnWeb (E1); ProxyStatus/Session model round-trips; backoff timing math (§7.3) Console ↔ real Go control plane (booted via containers submodule): list/create/revoke device, policy edit → effect-diff, WS event fold; auth token exchange against a real IdP/stub	run recording
e2e	full admin journey in a real browser (Playwright/integration_test web): login (OIDC PKCE + device-token) → device list → policy edit → see live route.changed → logout	window-scoped MP4 (§11.4.159)
full-automation (§11.4.25/.52/.98, deterministic §11.4.50)	the e2e journey re-runnable headless -count=3 identical, no human step; OCR-confirm the NetworkTile/audit values on screen (§11.4.158)	deterministic run logs + OCR verdict
Challenges (challenges submodule §11.4.27(B))	a Challenge bank entry: "open Console, edit a policy, observe live recompile within budget" scored on captured evidence	challenge result.json
HelixQA (helix_qa submodule)	written suites per Console screen + autonomous QA session driving the admin flows	HelixQA session report
build gate (no-FFI-on-web)	CM-WEB-NO-CORE-FFI asserts zero helix_core_ffidart:ffi in the web graph (§2.3); paired §1.1 mutation =	gate log + mutation diff

Test type (§11.4.169)	Web-Console coverage	Evidence
	add the import → gate FAILs	
DDoS/load-flood	WS reconnect storm: N tabs reconnect against the gateway; backoff jitter prevents thundering herd (§7.3)	latency/ connection- count chart
security (§11.4.10 + security submodule)	XSS cannot read access_token (memory- only) nor refresh_token (httpOnly); CSP blocks inline script; PKCE state CSRF; no token in localStorage; no PII in anonymous mode	security scan report
stress+chaos (§11.4.85)	kill the WS mid-session, drop network (onLine=false), bfcache freeze/thaw → Resyncing heals; corrupt/expire token mid-flight → re- auth; proxy carrier dropped → ProxyError honest	chaos recovery trace
concurrency/ atomicity	simultaneous REST mutation + WS event fold do not double-apply a device state; optimistic edit + server event reconcile	test logs
race-condition/ deadlock	token-refresh racing in- flight REST/WS calls (single-flight refresh, queued retries); no deadlock between auth + WS providers	race-probe logs
memory	Console bundle has no core_ffi weight (WI1); proxy WASM heap bounded, refuses-not- OOM (§8.7); soak the proxy and sample tab RSS	heap snapshot + RSS chart
benchmarking/ performance	first-paint time (deferred modules), %K palette latency, WS	perf trace

Test type (§11.4.169)	Web-Console coverage	Evidence
	event→render latency, proxy RTT vs direct	

Honest SKIPs (§11.4.3, never silent): the WASM-proxy soak + RTT tests are ProxyUnavailable-gated where a CI browser lacks WebTransport — recorded as a tracked migration item, never a faked PASS (§8.2 UNVERIFIED, §13 D-WEB-1).

Anti-bluff coupling (§11.4.107/§11.4.158). A green widget test is not proof an admin can manage the network. The **e2e + integration layers** are the evidence: a window-scoped MP4 of the real admin journey in a real browser against the real control plane, with the on-screen NetworkTile/audit values OCR-confirmed. The proxy's "on" claim is earned only by a captured request demonstrably traversing the gateway (aggregate counter delta), never by the chip color alone.

12. Phase → task → subtask plan

Aligned to the program roadmap [SYN §4]; each subtask is a workable item (§11.4.93 SQLite SSoT).

Phase 1 — Console MVP (Web build) [04_P1, 04_UI §8.3]

- **T1.W1 Web-flavor build manifest** — apps/console pubspec (no core_ffl), the §2.2 conditional-import seam, NoCore + NoTunnelPlatform, the CM-WEB-NO-CORE-FFI drift gate (§2.3).
- **T1.W2 Auth** — OIDC PKCE flow + anonymous device-token, the §6.3 session machine, the §6.4 token-storage policy, AuthController.
- **T1.W3 helix_api data layer** — generated REST client (auth-injected), the §7.2 WS/SSE event fold, the §7.3 reconnect state machine (bfcache/visibility/ onLine aware).
- **T1.W4 Console screens** — ConsoleShell + AdaptiveScaffold, Devices, Networks (NetworkTile), Policy (PolicyEditor effect-diff), Audit (live), responsive breakpoints, *K palette, a11y (WI4).

Phase 3 — WASM MASQUE proxy (opt-in, caveated) [04_P3, 03-CLIENT T3.3]

- **T3.W1 WASM spike (gate G-WEB)** — compile the helix-transport MASQUE subset to wasm32; prove a browser

WebTransport session reaches the gateway :443/udp MASQUE endpoint and round-trips one request. **Pass = a captured request traversing the gateway** (counter delta), else honest ProxyUnavailable (§8.2 D-WEB-1).

- **T3.W2 Proxy facade + fetch hook** — WasmMasqueProxy Dart facade (§8.3), page-scoped fetch interception, ProxyStatus chip (§8.4), the §8.5 honesty copy, §8.6 aggregate-only counters, §8.7 bounded heap.
- **T3.W3 Proxy soak (gate G-WEB-MEM)** — sustained transfer, sample tab RSS, refuse-not-OOM (§8.7); the browser analogue of G3.

13. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved.

#	Decision	Option A	Option B	Recommendation
D-WEB-1	Browser MASQUE carrier	WebTransport (HTTP/3 datagrams) drive the gateway CONNECT-UDP endpoint	fall back to WebSocket-tunnelled datagrams where WebTransport absent	A, gated on T3.W1 — prove the WebTransport↔MASQUE mapping (UNVERIFIED §8.2); where absent, honest ProxyUnavailable (never B-as-silent-fallback).
D-WEB-2	Token/cookie handling host	behind the gateway reverse proxy (cookie set by gateway)	Console talks the Go API directly (API sets cookie)	A — the gateway already fronts :443; keeps the httpOnly cookie same-site with the Console origin (§6.4 UNVERIFIED).
D-WEB-3	Web renderer	CanvasKit default	--wasm (Wasm GC) for heavy admin tables	A now, B when toolchain-stable — measure first-paint + table-scroll perf (§9, §11 benchmarking).
D-WEB-4	Ship the WASM proxy at all	Phase-3 opt-in, caveated	Console-only, no proxy ever	A — explicit browser-scoped value (CI3/WI5); it is never a device VPN [04_ARCH §5.7].

14. Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
Web-flavor manifest (helix_design+helix_api+helix_domain+helix_ll0n, no core_ffl/shim) + CM-WEB-NO-CORE-FFI gate	§2	apps/console, CI drift check
NoCore / NoTunnelPlatform null objects (throw UnsupportedOnWeb, never fake-success)	§3/§4	helix_domain, all flavors' web compile
Console session + auth lifecycle (OIDC PKCE + device-token, token-storage policy)	§6	doc 02 identity (/v1/auth/token), Console
WS/SSE reconnect state machine (bfcache/visibility/onLine aware, snapshot-heal)	§7.3	doc 02 event taxonomy, Console live views
WasmMasqueProxy contract + ProxyStatus (distinct from TunnelStatus; browser-scoped; aggregate-only)	§8	doc 01 gateway MASQUE edge, the proxy chip
No-system-tunnel-in-browser structural fact (WI2, §11.4.112)	§1	every Web claim; the user-facing copy

Sources verified

- 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md [04_UI] — §1 (Console = no core_ffl, only Web build), §2.console (capabilities {admin}), §4.3 (Console live WS fold), §7 (responsive breakpoints, ⌘K, deferred admin modules), §8.2 (OIDC + anonymous device-token), §8.3 (Console web+desktop), §9 (a11y, state announced not colored), §10.web (Wasm/CanvasKit, no core_ffl weight), §11 (web in the CI matrix).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH] — §2 (roles: Console = admin, Web responsive), §3.3 (MASQUE CONNECT-UDP/H3 = the QUIC mode), §5.2 (Flutter reaches Web via CanvasKit/Wasm), §5.6 (no Electron), §5.7 (**web cannot run a real device VPN; optional in-page WASM MASQUE proxy = browser- scoped, not a system tunnel**), §6 (REST + WS/SSE), §7 (no-logging).

- `final/03-client-core-and-ui.md` [03-CLIENT] — CI3 (Console = Web, `helix_api` only), §3 (FFI surface + TunnelStatus/Shields/ExitOption mirror), §5.7 (Web — no shim), §6 (flavor + capability gating, `runHelixApp`), §7 (design tokens, signature components), §12 (testing + FFI contract test).
- `final/v02-data-plane/orchestrator-and-state.md` [ORCH] — §4.1 (canonical 5-variant TunnelStatus), §4.5 (no per-conn records, aggregate only), §7.2 (jittered-exponential backoff shape reused by §7.3).
- `final/v02-data-plane/transport-trait.md` [TX-TRAIT] — the Transport trait
 - MASQUE path the WASM proxy compiles a subset of (§8.2).
- [05_YB0] operator mandate — eight-platform requirement incl. responsive Web. [SYN] cross-document synthesis — §1 (roles/apps), §2 (Web = admin/proxy only, no system TUN), §5 (Console = only Web build, no `core_ffi`), §7 (no-logging).
- [`research-flutter_ffi`] — Dart conditional-import / null-object pattern for excluding `dart:ffi` from a web target; SPA PKCE + `httpOnly-refresh-cookie` hardening. [`research-ios_android`] — native-vs-browser tunnel capability boundary (browsers have no TUN API).

*Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (decisions = options)

- recommendation; UNVERIFIED marks), §11.4.112 (structural-impossibility verdict for browser TUN), §11.4.5/§11.4.69/§11.4.107/§11.4.158/§11.4.159 (captured/ window-scoped evidence), §11.4.10 (credentials/token handling), §11.4.169 (comprehensive test-type coverage), §11.4.28/§11.4.74 (decoupled reusable packages), §11.4.76 (containers submodule for integration infra), §11.4.151/ §11.4.155 (release/recording prefixes), §11.4.156 (no active CI).*

End of Volume 4 (Clients) — Web Console (no tunnel).